Bubble Sort Rounds II

Time limit: 1.00 s **Memory limit:** 512 MB

Bubble sort is a sorting algorithm that consists of a number of rounds. On each round the algorithm scans the array from left to right and swaps any adjacent elements that are in the wrong order.

Given an array of n integers, find out the contents of the array after k bubble sort rounds.

Input

The first line has two integers n and k: the array size and the number of rounds.

The next line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print n integers: the contents of the array after k rounds.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $0 \le k \le 10^9$
- $1 \le x_i \le 10^9$

Example

Input	Output
5 2 3 2 4 1 4	2 1 3 4 4